Who Survives the Shock?

Global Crisis & Resilience Analysis Framework

Business proposal for global resilience analytics

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Executive Summary

Who Survives the Shock? is an advanced Global Crisis & Resilience Analysis Framework designed to measure and compare the resilience of 100 countries across six strategic domains: Economic Stability, Food Security, Healthcare Capacity, Digital Infrastructure, Climate & Energy, and Political Stability. This multidisciplinary DEPI Mega Project spans 24 years of historical data (2000–2023) and delivers end-to-end analytics solutions across five platforms — Excel, SQL, Python, Power BI, and Tableau.

By integrating data from the World Bank, FAO, and other global development sources, the framework transforms raw indicators into actionable intelligence that supports data-driven decision-making, strategic planning, and risk assessment. The project delivers a Composite Global Resilience Index, interactive multi-platform dashboards, machine learning-based country estimation, and 2030 resilience forecasting with optimistic, base, and pessimistic scenarios.

Problem Statement

Global crises rarely occur in isolation. Economic instability, food insecurity, healthcare challenges, climate pressures, energy vulnerability, and political instability are deeply interconnected forces that together determine a country's ability to withstand and recover from shocks. Traditional development metrics focus on single dimensions — economic growth, healthcare performance — and fail to capture this multi-domain complexity.

Without a unified analytical framework, policymakers and researchers lack the tools to identify structural vulnerability, benchmark regional performance, or anticipate future resilience trajectories. This project was designed to bridge that gap by building a comprehensive, data-driven resilience assessment system that answers one critical question: which countries are merely surviving global shocks, and which are truly resilient?

Project Objectives

- **Multi-Domain Resilience Measurement:** Analyze global datasets to construct a composite resilience index covering economic, food, healthcare, climate, digital, and governance dimensions.
- **Dashboard Integration:** Create interactive dashboards across Excel, Power BI, and Tableau for improved decision-making and storytelling.
- **Country & Regional Benchmarking:** Rank and compare 100 countries and 10 global regions based on resilience performance from 2000 to 2023.
- **Vulnerability Identification:** Detect countries and regions facing the highest risk across multiple resilience domains.
- **Predictive Analytics:** Apply machine learning to estimate resilience for countries with incomplete data and forecast 2030 resilience trajectories.
- **Actionable Insights:** Generate evidence-based recommendations to support strategic planning and data-driven policymaking.

Team Structure

Name	Role	Responsibility
Eslam Hassan	Team Lead	Project Management & Data Analysis
Mostafa ELrkhawy	Data Analyst	Research & Data Analysis
Habiba Ahmed	Data Analyst	Research & Data Analysis
Sara Mohamed	Data Analyst	Research & Data Analysis
Ann Osama	Data Analyst	Research & Data Analysis
Omar Mahmoud	Data Analyst	Research & Data Analysis

The team operates in a cross-functional model to ensure comprehensive coverage of all analytical domains, from raw data processing through to final dashboard delivery and stakeholder presentation.

Project Workflow

1. Data Collection from World Bank and FAO sources
2. Data Cleaning, Normalization, and Feature Engineering
3. Galaxy Schema Data Modeling and Warehousing
4. Domain Score Calculation and Composite Resilience Index Construction
5. Exploratory Data Analysis (EDA) and Statistical Insights
6. Multi-Platform Dashboard Development (Excel, SQL, Python, Power BI, Tableau)
7. Machine Learning Model Training for Estimated Countries and 2030 Forecasting
8. Scenario Analysis and Executive Recommendations

Data Analysis Implementation

The project was implemented independently across five analytics platforms, each covering the complete analytics lifecycle from raw data to insights and recommendations.

Implementation	Timeline	Scope
Excel Analytics	Week 1	End-to-End Analytics Solution
SQL Data Warehouse	Week 2	End-to-End Analytics Solution
Python + Machine Learning	Week 3	Analytics & Predictive Analytics Solution
Power BI Dashboard	Week 4	End-to-End Analytics Solution
Tableau Storytelling	Week 5	Interactive Analytics & Storytelling Solution

Excel Implementation: Data cleaning, transformation, Power Query, Power Pivot, KPI development, interactive dashboard design, and business insights.

SQL Implementation: Data cleaning, Galaxy Schema design, data warehousing, analytical queries, resilience scoring, risk assessment, and recommendations.

Python Implementation: EDA, statistical analysis, time-series analysis, correlation analysis, composite resilience index construction, machine learning model development, feature importance analysis, country estimation, 2030 forecasting, multi-scenario analysis, and interactive Streamlit dashboard development.

Power BI Implementation: Power Query ETL, data modeling, DAX measures, KPI development, interactive dashboards, storytelling, and executive insights.

Tableau Implementation: Interactive dashboards, storytelling, KPI visualization, risk analysis, and business insights.

Data Sources & Data Model

World Bank Indicators

- Fixed Broadband Subscriptions
- Internet Users
- GDP Growth
- Inflation
- Food Imports
- Prevalence of Undernourishment
- Health Expenditure
- Hospital Beds
- Physicians
- Political Stability
- Access to Electricity
- Access to Clean Fuel
- CO_2 Emissions
- Renewable Energy
- Electricity Consumption

FAO Food Price Index

- Food Price Index
- Dairy Index
- Cereals Index
- Oils Index
- Meat Index
- Sugar Index

Galaxy Schema Data Model

The project follows a dimensional modeling approach using a Galaxy Schema with two fact tables and four dimension tables.

Table	Type	Contents
Fact_Global_Indicators	Fact	Indicator values, normalized scores, resilience metrics, country/indicator/year keys
Fact_Food_Index	Fact	Food price index, commodity categories, monthly trends, price metrics
Dim_Country	Dimension	Country name, country code, region
Dim_Year	Dimension	Year, decade

Dim_Indicator	Dimension	Indicator name, domain classification, unit
Dim_Type	Dimension	Food commodity category

Dashboard & Visualization Track

The visualization layer of this project delivers seven specialized dashboards covering all resilience domains. Each dashboard is designed to support storytelling and provide decision-makers with a clear view of global risks and resilience patterns.

Global Overview Dashboard

KPIs: Resilience Champion, Critical Crisis Zone, Global Inflation Rate, Global GDP Growth. Visualizations include a Regional Distribution Analysis, Average Regional Stability Score, Global Composite Score Map, and GDP Growth vs. Inflation Scatter Plot.

Economic Fragility Dashboard

Analyzes inflation, GDP growth, and government debt to evaluate macroeconomic stability and identify regions most vulnerable to economic shocks.

Food Security Dashboard

Examines food price volatility, hunger levels, and food import dependency to map global food insecurity and nutritional vulnerability.

Digital Infrastructure Dashboard

Measures internet penetration, broadband density, and connectivity gaps to highlight the digital divide between regions and countries.

Energy & Climate Dashboard

Explores carbon footprint, renewable energy adoption, electricity access, and climate readiness to assess sustainability performance.

Healthcare Dashboard

Assesses health expenditure, medical capacity, and physician availability to evaluate national healthcare readiness against crises.

Political Stability Dashboard

Evaluates stability trends, political volatility, and governance quality and their influence on long-term resilience outcomes.

Tools & Technologies

Track	Tools Used
Data Warehousing	MySQL, SQL Server, Galaxy Schema Design
Analytics & EDA	Python, Pandas, NumPy, Plotly, Scikit-Learn, Streamlit
Business Intelligence	Power BI, DAX, Power Query
Storytelling	Tableau Public, Tableau Prep
Spreadsheet Analytics	Excel, Power Query, Power Pivot

Use Case Scenarios

Scenario 1: Government policymakers use the Composite Resilience Index dashboards to benchmark their country against regional peers and identify priority domains for strategic investment.

Scenario 2: Development organizations and NGOs use the Risk Assessment module to identify countries in critical conditions and allocate emergency resources effectively.

Scenario 3: Researchers and academics use the 2030 Forecasting Engine with multi-scenario analysis to model the impact of policy decisions on future resilience trajectories.

Scenario 4: International organizations use the Estimated Country Explorer to assess resilience for countries with incomplete data, ensuring no country is left out of global analysis.

Key Performance Indicators (KPIs)

- Composite Resilience Score across 100 countries, benchmarked annually from 2000 to 2023.
- Domain scores for all six resilience dimensions: Economic, Food Security, Healthcare, Digital Infrastructure, Climate & Energy, and Political Stability.
- Global Resilience Tier classification (Resilient, Stable, Vulnerable, At-Risk, Critical) for each country and year.
- Regional performance ranking covering 10 global regions.
- Machine learning model accuracy for country estimation and 2030 forecasting.
- 2030 scenario forecast under Optimistic, Base, and Pessimistic conditions for all tracked countries.

Implementation Timeline

Week	Activities
Week 1	Data ingestion from World Bank and FAO; Excel analytics implementation; Power Query ETL and Power Pivot modeling
Week 2	Galaxy Schema design; SQL data warehouse construction; resilience scoring queries and risk assessment
Week 3	Python EDA and statistical analysis; composite resilience index; ML model development and Streamlit dashboard
Week 4	Power BI dashboard development; DAX measures; KPI design and interactive storytelling
Week 5	Tableau dashboard development; final report preparation; client engagement and presentation

Expected Outcomes

- A clean, unified HR dataset integrated across 100 countries and 24 years, ready for multi-platform analytics.
- End-to-end analytics solutions deployed across Excel, SQL, Python, Power BI, and Tableau.
- A Composite Global Resilience Index enabling structured country and regional benchmarking.
- Machine learning-based resilience estimation for countries with incomplete historical data.
- A 2030 forecasting engine delivering optimistic, base, and pessimistic resilience scenarios.
- Actionable strategic recommendations to support policymakers, researchers, and development organizations.

Challenges and Mitigation Strategies

Challenge	Solution
Inconsistent and Missing Data Records	Cross-validation using SQL and statistical imputation techniques
Multi-Source Data Integration	Unified Galaxy Schema with surrogate keys and domain classification
Indicator Scale Differences	Min-Max normalization with direction adjustment for inverse indicators
Machine Learning Model Overfitting	Cross-validation, hyperparameter tuning, and feature importance analysis
Time Management Across Five Platforms	Weekly milestones with dedicated implementation tracks per tool

Conclusion & Future Enhancements

The Global Crisis & Resilience Analysis Framework delivers a comprehensive, multi-platform understanding of how countries withstand and recover from global shocks. By integrating economic, food security, healthcare, digital infrastructure, climate, and political stability data into a unified composite index, the project provides a structured and reproducible methodology for resilience assessment at a global scale.

The multi-tool implementation — spanning Excel, SQL, Python, Power BI, and Tableau — demonstrates that the same analytical challenge can be addressed through different technology stacks, each offering unique capabilities. The addition of machine learning for country estimation and 2030 forecasting elevates the project beyond traditional analytics, providing predictive intelligence that supports forward-looking strategic planning.

Recommendations

- Accelerate digital infrastructure investment in low-connectivity regions to reduce structural resilience gaps.
- Strengthen healthcare capacity and expand medical workforce access in vulnerable countries.
- Reduce food import dependency through agricultural diversification and domestic supply chain development.
- Expand renewable energy adoption to improve energy resilience and reduce environmental vulnerability.
- Strengthen governance frameworks and institutional stability as leading indicators of long-term resilience.

Future Vision

- Expand coverage to over 180 countries with real-time data integration.
- Launch an open-access SaaS resilience monitoring platform.
- Integrate AI-powered alerts for early warning of emerging resilience risks.
- Develop mobile dashboard applications for field researchers and policymakers.

Key Findings & Business Insights

Overview & Domain Insights

- Europe demonstrates the strongest and most consistent resilience performance across all domains.
- Digital Infrastructure is the fastest-improving global resilience domain over the 2000–2023 period.
- Healthcare remains the weakest global domain, with persistent gaps in medical capacity and expenditure.
- Political Stability has shown a declining trend since 2010, creating systemic risk across multiple regions.
- Regional resilience differences are structural rather than temporary, requiring long-term policy intervention.

Risk & Food Security Insights

- Inflation is one of the strongest single drivers of vulnerability and composite score decline.
- Food insecurity amplifies socio-economic risk and frequently precedes healthcare system deterioration.
- Political instability acts as a leading indicator of future economic fragility.
- Africa faces significant multi-domain challenges with compounding healthcare and food insecurity risks.
- The 2022 food crisis created unprecedented global pressure on food price indexes across all categories.

Statistical Insights

- The digital divide is structural rather than gradual and reflects deeper development inequalities.
- Energy poverty and food insecurity are strongly interconnected across low-income regions.
- Multiple indicators exhibit non-normal distributions, requiring robust statistical methods for analysis.
- Inflation outliers create significant systemic risk that standard parametric assumptions fail to capture.

Recommendations

Accelerate Digital Inclusion: Invest in broadband and internet infrastructure to strengthen resilience in digitally excluded regions.

Strengthen Healthcare Systems: Expand healthcare capacity, grow medical workforces, and improve access to essential health services globally.

Improve Food Security: Reduce import dependency, diversify food supply chains, and strengthen domestic agricultural production.

Expand Sustainable Energy Access: Increase renewable energy adoption and electricity accessibility to simultaneously enhance economic and environmental resilience.

Enhance Governance & Stability: Strengthen institutional frameworks and governance to improve political stability as a leading resilience factor.

Promote Data-Driven Decision Making: Leverage integrated resilience monitoring systems to identify emerging risks early and allocate resources more effectively across regions.